FINAL YEAR PROJECT PROPOSAL

AI-Powered Sharia Inheritance Distribution System

A decision-support application for calculating and explaining Islamic inheritance shares

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Note: This proposal describes a prototype for educational and decision-support use. Inheritance outcomes can depend on the applicable school of jurisprudence, family circumstances, debts, wills and local law. The system must show its assumptions and should not replace review by a qualified scholar or legal professional.

1. Introduction

1.1 Background

Islamic inheritance (fara'id) provides principles for distributing a deceased person's estate among eligible heirs. The calculation can involve several relatives, fixed shares, residuary heirs, exclusions and estate-related obligations. Families may find the process difficult to understand, particularly when several potential heirs are involved or when the information available is incomplete.

1.2 Project Overview

The proposed project is a web-based AI-powered Sharia inheritance distribution system. A user will enter relevant information about the deceased, the estate and surviving relatives. A transparent calculation engine will apply a defined set of inheritance rules, while AI-assisted features may help organize user input, explain terms and present the result in accessible language. The application will display the calculated shares, assumptions and a breakdown of how the result was reached.

1.3 Project Purpose

The purpose is to develop a manageable student-level application that demonstrates how software, rule-based computation and carefully limited AI assistance can make inheritance calculations easier to follow. The system is intended to support understanding and preliminary calculation, not to issue an independent religious or legal ruling.

2. Problem Statement

Inheritance calculations can be challenging for people who are unfamiliar with fara'id rules. Manual calculations may be time-consuming and can lead to mistakes when the family structure, eligible heirs, estate value or relevant obligations are recorded incorrectly. Some digital calculators provide only a final amount without clearly explaining the assumptions or the calculation steps. There is a need for a user-friendly system that gathers structured family information, calculates shares consistently under a clearly selected rule set, explains the result and flags cases that require expert review.

3. Proposed Solution

The project will develop a web application in which users enter the deceased person's family relationships and the distributable estate amount. The system will validate required fields, identify possible heirs and calculate shares using an explicit, testable rules engine based on the selected and documented inheritance framework. AI features may assist with guided data entry and plain-language explanations, but the share calculation itself should remain rule-based and auditable rather than relying on a generative model to invent fractions.

Before distribution is calculated, the application will prompt users to account for relevant estate obligations, such as funeral expenses, debts and valid bequests, according to the framework configured for the prototype. The result will show each included heir, the share or fraction, the amount, and explanatory notes. Ambiguous family situations, conflicting information or unsupported cases will be flagged for qualified review instead of being presented as certain.

4. Project Objectives

4.1 General Objective

To design and develop an AI-assisted Sharia inheritance distribution application that calculates and explains potential heirs' shares using a transparent, documented rules engine.

4.2 Specific Objectives

- To create a simple interface for entering the deceased person's details, estate value and family relationships.
- To validate user input and identify missing or inconsistent information before calculation.
- To implement a rule-based calculation module for a clearly defined and documented set of inheritance cases.
- To show each eligible heir's calculated fraction and estimated monetary amount, along with the calculation steps.

• To use AI assistance, where appropriate, for guided input and plain-language explanations without allowing it to override the calculation rules.

• To flag unsupported, ambiguous or exceptional cases for review by a qualified scholar or legal professional.

• To store calculation cases securely and provide a basic summary or printable report.

• To evaluate calculation correctness against verified sample cases and document limitations.

5. Scope of the Project

5.1 In Scope

• User entry of estate value and relevant family relationships.

• Input validation and clear prompts for missing information.

• A documented, rule-based inheritance calculation module for the selected prototype cases.

• Calculation results showing shares, amounts and a step-by-step explanation.

• AI-assisted help text or plain-language summaries, subject to validation.

• Basic case history and report display, with access controls.

• Testing against reference cases reviewed by a knowledgeable subject-matter expert.

5.2 Out of Scope

• Replacing a mufti, qualified Islamic inheritance specialist, lawyer or court.

• Claiming to cover every school of jurisprudence, every exceptional case or every jurisdiction.

• Automatically resolving disputed family relationships, ownership claims, wills or legal documents.

• Executing bank transfers, changing land ownership or distributing real assets.

• Providing a binding legal or religious ruling without human review.

5.3 Possible Future Enhancements

Future versions could support multiple documented jurisprudential rule sets, multilingual guidance, verified expert review workflows, document-assisted data entry and additional reporting. Such extensions would require expert validation, privacy safeguards and supervisor approval.

6. Target Users

The intended users are individuals or families seeking to understand a preliminary inheritance calculation, students studying fara'id, and advisers who want a structured calculation aid. An administrator may maintain the supported rule set and test cases. The initial prototype will be designed for controlled demonstrations and educational use rather than unrestricted public reliance.

7. Proposed Features

7.1 Case and Family Information

The user will enter the estate amount and relevant family relationships, such as spouse, parents, children and other relatives supported by the prototype. The interface will explain which information is required and will ask the user to confirm the family tree before proceeding.

7.2 Input Validation

The system will check for missing fields, impossible or conflicting entries, and amounts that are not valid numbers. It will display clear messages and request correction rather than silently assuming missing facts.

7.3 Inheritance Calculation Engine

A deterministic rules module will apply the documented rules for the supported cases. Rules, assumptions and test cases will be kept separate from the interface so that results can be tested and reviewed. Unsupported cases will be marked for expert review.

7.4 AI-Assisted Explanation

AI may translate technical calculation steps into plain language, explain terminology and guide users through the form. Any explanation will be generated from the calculation engine's structured output, and the system will not allow AI-generated text to change the numeric shares.

7.5 Distribution Summary

The result page will show the net estate amount used in the calculation, each eligible heir, the calculated fraction or share, the estimated amount and explanatory notes. The system will identify rounding and any assumptions used.

7.6 Case History and Reports

If time permits, users will be able to save a case and print or export a summary. Sensitive family and financial details will be limited to what is necessary and protected by access controls.

7.7 Expert Review Warnings

Cases involving uncertainty, disputed facts, unusual relationships, jurisdiction-specific legal issues or rules outside the prototype will display a clear warning that professional review is required.

8. Proposed Technologies

Technology	Planned Use
Python	Main programming language and calculation logic.
Flask	Lightweight web application framework.
SQLite	Local database for case records and supported rule/test metadata.
HTML, CSS and Bootstrap	Web forms, result pages and responsive interface.
Rule-based engine	Transparent and repeatable calculation of supported inheritance cases.
AI / NLP component	Optional guided input and plain-language explanations based on verified outputs.
NumPy (if required)	Numerical operations and validation support.
Git and GitHub	Version control and project code management.

The final technology stack may be adjusted after an initial feasibility test and supervisor feedback. Any AI service used must be assessed for privacy, reliability and cost.

9. Expected Outcomes

The expected outcome is a working prototype that accepts structured inheritance information, calculates supported cases using documented rules, and presents an understandable distribution summary.

- A Python-based web application with an inheritance case input form.
- A tested calculation engine for a clearly stated subset of cases.
- A result page showing heirs, fractions, amounts and explanations.
- AI-assisted guidance or summaries, if feasible and validated.
- A small set of expert-checked test cases and recorded evaluation results.
- Project source code, user instructions and documentation of limitations.

10. Project Significance

The project provides practical experience in software engineering, rule-based reasoning, web development, data validation and responsible AI integration. It addresses a meaningful real-world information need by making calculations easier to inspect and understand. Its emphasis on transparent rules and human review also demonstrates why high-impact religious, legal and financial decisions should not depend on an opaque AI answer alone.

11. Feasibility and Ethical Considerations

11.1 Technical Feasibility

A prototype can be built with Python, Flask and SQLite on a standard laptop. A limited, well-defined set of inheritance scenarios will keep implementation and testing manageable. The calculation rules should be documented before coding and validated against reliable references and subject-matter review.

11.2 Schedule Feasibility

The work will proceed in stages: requirements and rule selection, interface design, calculation engine, explanation layer, database integration, testing and documentation. A working calculation core will be prioritized before optional AI enhancements.

11.3 Religious, Legal and Ethical Considerations

Inheritance shares can depend on detailed facts and on the applicable jurisprudential and legal framework. The system must state which framework it implements and must not imply that one configuration represents every interpretation. A qualified subject-matter expert should review the rules and test cases. Users should be told that outputs are preliminary decision support, not a binding fatwa or legal determination.

Family relationships and estate values are sensitive data. The prototype should collect only necessary information, restrict access, avoid publishing real case data in a public repository and provide a way to correct errors. AI explanations must be grounded in the calculation output, and uncertainty should be communicated clearly.

11.4 Limitations

The system’s reliability will be limited by the accuracy of the user-provided family information, the rule set implemented and the cases used for testing. It may not support exceptional cases or local legal requirements. AI-generated explanations may be unclear or inaccurate and therefore require guardrails and review. These limitations will be stated in the interface and final report.

12. Development Methodology

• Requirements and rule selection: define supported relationships, assumptions and the intended jurisprudential framework with supervisor and expert input.

• System design: prepare user flows, data structures, rule representation and calculation-result format.

• Calculation prototype: implement the supported cases as deterministic functions and test each rule.

• Interface and data validation: build forms that capture family details and prevent incomplete or invalid submissions.

• AI-assisted explanation: add optional plain-language explanations generated from verified structured results.

• Database and reporting: store permitted case data and create a clear distribution summary.

• Testing and evaluation: compare outputs against independently verified reference cases and test invalid or unsupported inputs.

• Documentation and demonstration: record the supported scope, results, limitations and instructions.

13. Tentative Project Timeline

Weeks	Activity	Deliverable
1–2	Confirm requirements, scope and expert-reviewed rule framework.	Approved scope and initial test cases.
3–4	Design data model, interface and calculation approach.	Design and input forms.
5–7	Implement rule-based calculation engine.	Initial working calculations.
8–9	Build validation, result breakdown and reporting.	Integrated calculation flow.
10–11	Add optional AI-assisted guidance and explanations.	Explanation prototype with guardrails.
12–13	Integrate database and improve user interface.	Working web prototype.
14–15	Run verification, usability and edge-case tests.	Evaluation results and fixes.
16	Complete documentation and presentation.	Final FYP submission.

This is a tentative 16-week plan and may be adjusted according to the university calendar, supervisor feedback and availability of subject-matter review.

14. Evaluation Plan

The system will be evaluated using a small set of inheritance scenarios whose expected outcomes have been independently checked by a qualified reviewer. The evaluation will measure correctness within the stated scope rather than claim universal coverage.

• Calculation correctness: compare each heir’s fraction and amount with the verified reference result.

• Completeness: check whether the system identifies missing information and flags unsupported cases.

• Input validation: test invalid values, inconsistent family information and incomplete forms.

• Explanation quality: review whether summaries accurately reflect the calculation engine and are understandable to test users.

• Usability: ask a small number of users to complete a sample case and record difficulties.

• Privacy and safety: verify access controls, data minimization and warnings for professional review.

Possible metrics include the proportion of supported test cases calculated correctly, number of invalid cases properly flagged, and user feedback on clarity. The final report will state the number and limits of the test cases.

15. References

1. The Qur’an, Surah An-Nisa (4:11–12 and 4:176), concerning inheritance shares. The relevant interpretation and application must be confirmed with a qualified scholar for the chosen framework.

2. Python Software Foundation. Python Documentation. https://docs.python.org/3/

3. Pallets Projects. Flask Documentation. https://flask.palletsprojects.com/

4. SQLite. SQLite Documentation. https://www.sqlite.org/docs.html

5. World Wide Web Consortium (W3C). Web Content Accessibility Guidelines (WCAG). https://www.w3.org/WAI/standards-guidelines/wcag/

6. Add the exact scholarly works, authoritative fara’id references, legal sources and AI documentation consulted during implementation, using the reference style required by the department.

16. Conclusion

The proposed AI-Powered Sharia Inheritance Distribution System aims to make preliminary inheritance calculations more structured, transparent and understandable. It will combine a documented rule-based calculation engine with a web interface and, if feasible, AI-assisted guidance. By displaying assumptions, calculation steps and review warnings, the prototype will prioritize traceability and responsible use. Its results will remain limited to the rules and cases that have been validated, and complex or disputed matters will be referred to qualified human experts.